INFO 7610 Topic Nat Lang Engr Meth/Tools

Data is All You Need

Agenda

- NLP Data
  - General Data
  - NLP data overview
  - Data collection and acquisition
  - Data annotation and labeling
  - Preprocessing and cleaning
  - Validation and quality
  - Ethics and Bias
- Present the Data
  - How to present the data
  - Language Models
- Discussion/ Q&A

What is Data

What is Data

- **Data** - are encodings that represent the qualitative or quantitative attributes of a variable or set of variables.
- **Data** (plural of "datum", which is seldom used) – are typically the results of measurements, computations, or observations and can be the basis of graphs, images of a set of variables.
- **Data** - are *often* viewed as the lowest level of abstraction from which information and knowledge are derived

DIKW Model

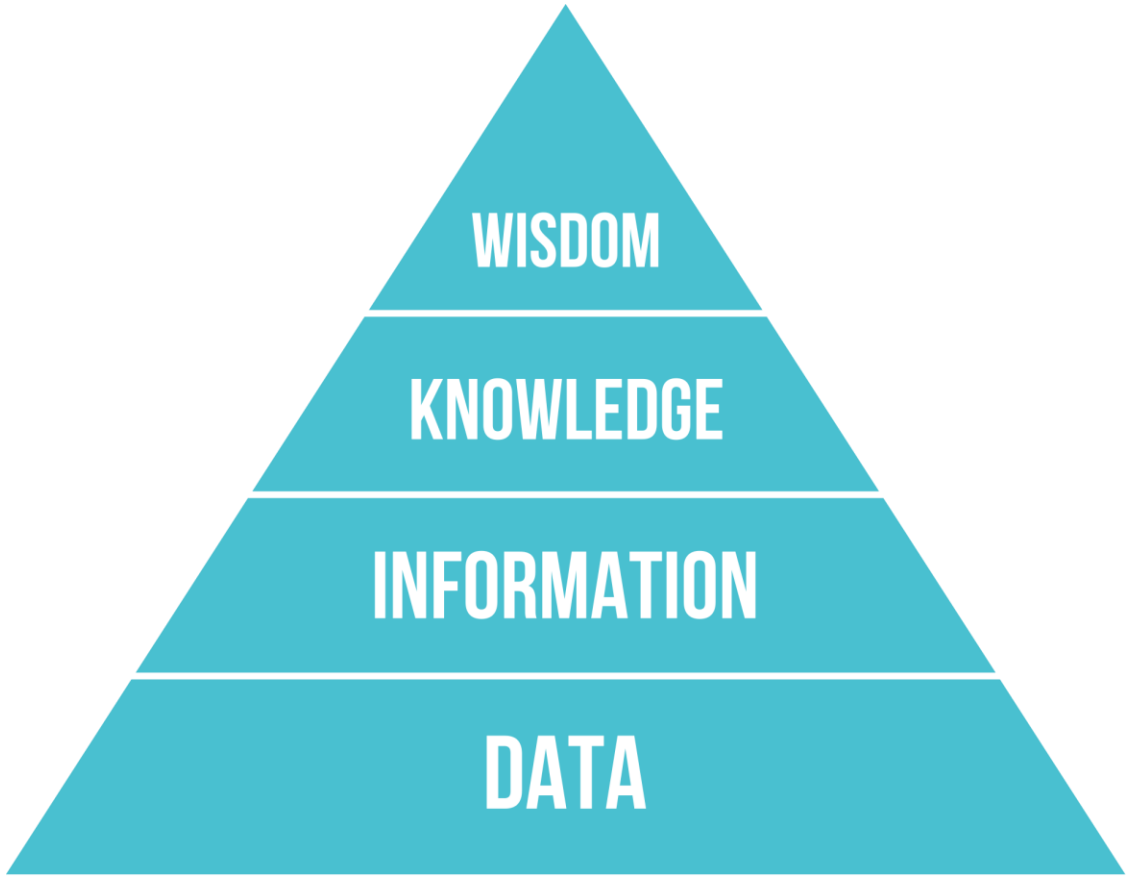


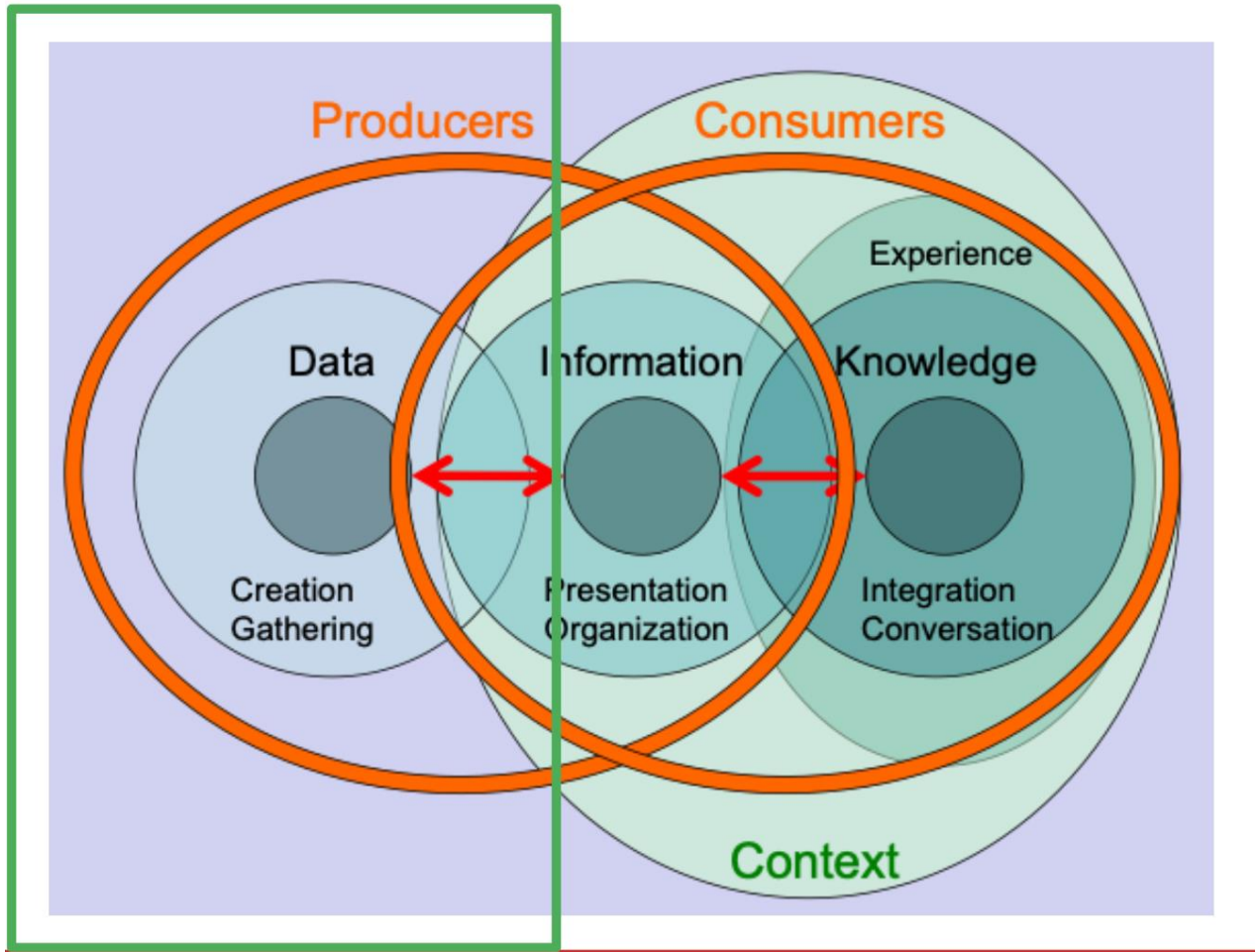
Image Source: https://en.wikipedia.org/wiki/DIKW_pyramid

Data is All you Need

- **Data as the foundation**
  - ML models are only as good as the data they're trained on.
- **Scale and complexity**
  - Data size can be large (internet-scale corpora) and complex (multilingual, domain-specific).
- **Diversity of tasks**
  - Classification, translation, QA, generation, etc. each require different data considerations.

Data Life Cycle

- Data-Information-Knowledge Ecosystem



Data Life Cycle

- **Collection** (where data comes from)
- **Cleaning & Curation** (ensuring data integrity, consistency)
- **Annotation** (labeling data for supervised tasks)
- **Preprocessing** (tokenization, normalization, etc.)
- **Split & Sampling** (train/validation/test sets)
- **Monitoring & Maintenance** (data drift, updates)

Common NLP Data Sources

- **Web Text**
  - Wikipedia dumps, Common Crawl, Reddit, Twitter
- **Curated Corpora**
  - Project Gutenberg, OpenSubtitles, newswire data
- **Proprietary/Industry Data**
  - Company logs, conversation transcripts, domain specific data, Catalog for ecommerce, etc
- **Benchmarks & Shared Tasks**
  - GLUE, SQuAD, CoNLL datasets

Data Formats & Storage

- **Structured vs. Unstructured v.s Semi-Structured**
- **Annotation Formats**
  - CoNLL format, BIO tagging, XML-based annotation
- **Data Versioning & Storage Tools**
  - Git LFS, DVC, Hugging Face Datasets

Question to Discuss

- Given a new NLP task: *Detecting offensive language in social media posts*, how would you go about collecting relevant data?

Data Annotation

- A painful process…
- **Why Annotation Matters**
  - Enables supervised and semi-supervised learning.
- **Types of Annotations**
  - Classification Labels (e.g., sentiment)
  - Sequence Labeling (e.g., NER, POS tagging)
  - Structural Annotation (syntax trees, dependency graphs)
  - Semantic Roles or Frames

Annotation Tools & Practices

- **Annotation Guidelines**:
  - Clear, consistent definitions for labels.
  - Example-based instructions.
  - Inter-annotator agreement as a quality metric (Cohen's kappa, Krippendorff's alpha).
- Level of Accuracy
  - Golden
  - Silver (?)
- Manual Annotation

Annotation Tools & Practices

- **Popular Tools**:
  - Prodigy
  - tagtog
  - LightTag
- LLM is here to help!

Data Preprocessing & Cleaning

- **Tokenization**: WordPiece, Byte-Pair Encoding (BPE), spaCy, NLTK, etc.
- **Lowercasing, Stemming/Lemmatization**: Task-dependent decisions.
- **Removing Noise**: HTML tags, special characters, repeated punctuation.
- **Handling Missing or Corrupt Data**: Dropping or imputing, if relevant.

Splitting the Dataset

- **Train/Validation/Test Splits**
  - Typical ratios (70/15/15 or 80/5/15).
- **Stratified Sampling**
  - Maintain label distribution.
- **Cross-Validation**
  - For limited data scenarios.
- **Temporal Splits**
  - For time-sensitive data (e.g., social media, news).

Data Quality Assessment & Bias

- **Bias in Data**: Representation bias, annotation bias.
- **Detecting Bias**:
  - Exploratory data analysis (demographics, topics).
  - Model-based diagnostics (performance disparities).
- **Mitigating Bias**:
  - Collecting diverse data.
  - Debiasing methods (adversarial training, domain adaptation).

Key Dimensions of Data Quality

- **Accuracy**:
  - Example: A person’s age recorded as 150 years old is inaccurate.
- **Completeness**:
  - Example: A customer record missing the address field can be incomplete.
- **Consistency**:
  - Example: The same product listed with different prices in two different databases.
- **Timeliness**:
  - Example: Stock market data from last year will not reflect current market trends.
- **Relevance**:
  - Example: Including irrelevant columns like "employee ID" in a customer satisfaction survey dataset.
- **Validity**:
  - Example: A phone number field containing alphabetical characters.
- **Uniqueness**:
  - Example: A customer record appearing twice in a database.

Methods of Data Quality Assessment

- **Descriptive Statistics**: Calculate means, medians, variances, and distributions to check data for outliers and inconsistencies.
- **Data Profiling**: Examining data to summarize its structure, content, and relationships to identify quality issues.
- **Data Auditing**: Conducting periodic checks and audits of data for adherence to quality standards.
  - Example: data drifting
- **Validation Rules**: Applying predefined rules to check if the data falls within acceptable ranges or formats.

Types of Bias in Data

- **Sampling Bias**:
  - Example: A survey conducted only in urban areas may not represent rural populations.
- **Measurement Bias**:
  - Example: A biased survey question might lead participants toward a particular response.
- **Label Bias**:
  - Example: In a sentiment analysis task, mislabeling tweets as positive or negative due to subjective human judgments.
- **Prejudicial Bias**:
  - Example: A facial recognition system trained predominantly on lighter-skinned faces may perform poorly on darker-skinned individuals.
- **Confirmation Bias**:
  - Example: Using only data that supports a specific theory while ignoring contradicting evidence.
- **Exclusion Bias**:
  - Example: Excluding people over the age of 65 from a medical study could lead to biased health outcomes.

Ethical & Legal Considerations

- **General Data Protection Regulation & Data Privacy**
  - Implications for storing user data.
- **Copyright & Licensing**
  - Using third-party texts.
- **Consent & Transparency**
  - In crowdsourced annotation.
- **Open-Source vs. Proprietary**
  - Restrictions on model sharing.

Tools & Libraries for NLP Data Handling

- **Python Ecosystem**: pandas, NumPy for data manipulation
- **NLP Libraries**: NLTK, spaCy, Hugging Face Transformers & Datasets
- **Annotation Tools**
- **Storage & Versioning:** Git

Summary & Key Takeaways

- **Data is central** to all NLP model building and evaluation.
- **Annotation quality** drives performance in supervised tasks.
- **Preprocessing** must be carefully designed for the domain and task.
- **Data biases** can significantly impact fairness and model reliability.
- **Continual iteration** is necessary (data collection → annotation → feedback → improvement).

Part II Language Models

Let the machine/model understand the data

Recap: Language Model

- p(I read a book about dogs)
  - How to assign a probability to a sentence?
- another view: distribution over next word:
  - p(dogs | I read a book about _____)

Recap: Language Model

Language Models: Definition

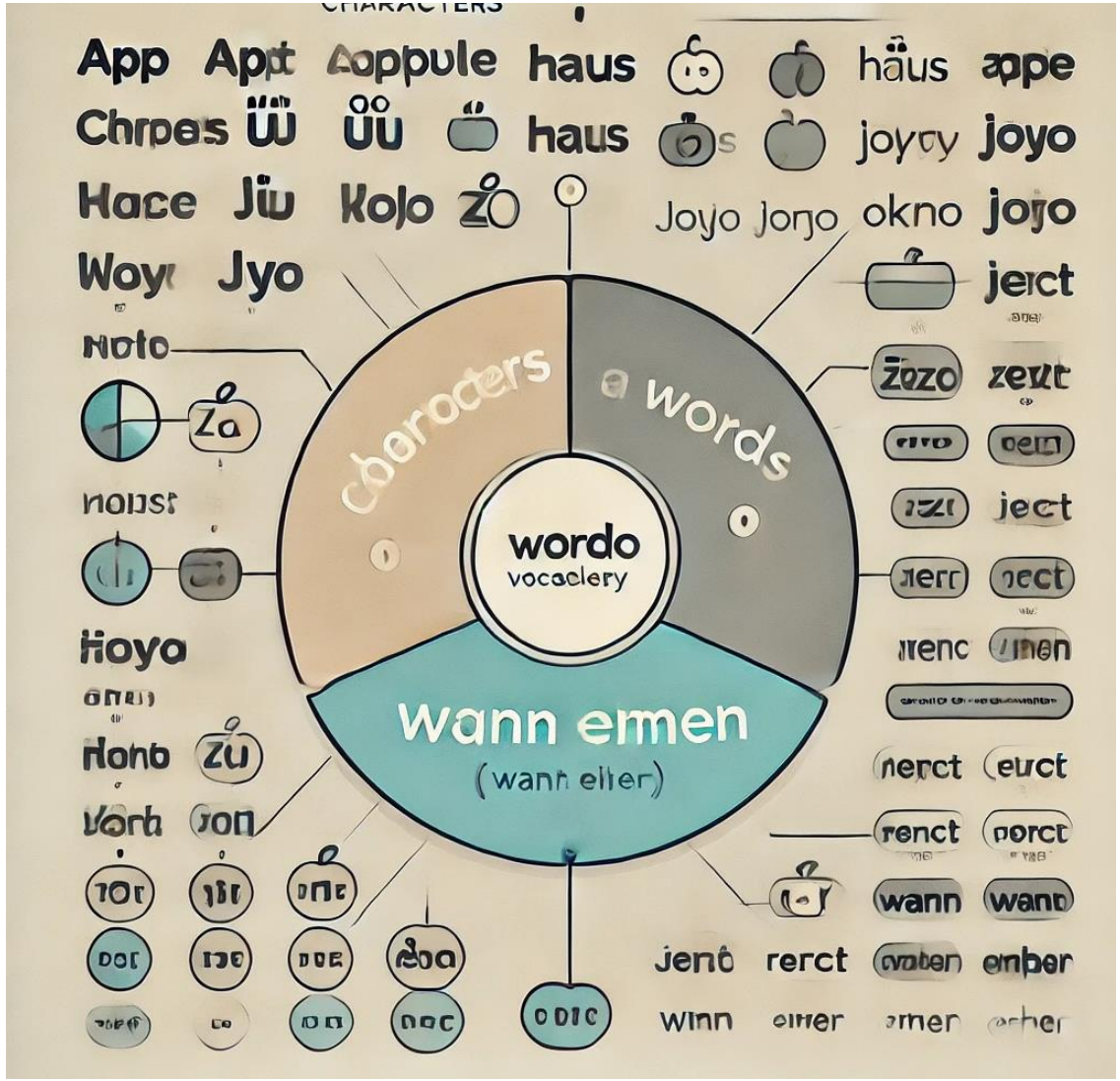
- A language model (LM) $p_{\boldsymbol{\theta}}$ is a **probability distribution over strings**
  - should tell us how likely a piece of text is to occur in a particular natural language domain (i.e., the domain of the training corpus)

$p_{\boldsymbol{\theta}} \rightarrow$

0.001	My dog took me for a walk.
....	....
....	....
....	....
0.1	I took my dog for a walk.

Language Model

- We define a string Y as a sequence of tokens
  - <y1,y2,y3,…>
- All y must be in the model's vocabulary V. A vocabulary is a set of symbols. These can be characters, words, subwords…Vocabulary units can also cross word-boundaries



Vocabulary

Aspect	Characters	Words	Subwords
Granularity	Smallest	Largest	Intermediate
Vocabulary Size	Small	Large	Medium
Flexibility	High (handles any text)	Low (OOV issues)	High
Computation	Slow (long sequences)	Efficient for short text	Efficient overall

Vocabulary should always include an EOS symbol, designating when the string is finished. Otherwise, we cannot globally normalize $p(\theta)$.

Building Blocks: How to Represent Language?

- In Neural Network

functions from vectors
to vectors

functions from vectors
to probabilities

(these are still functions from vectors to vectors)

Word Embedding

- **Idea:** learn an embedding from words into vectors
- Need to have a function W(word) that returns a vector encoding that word.

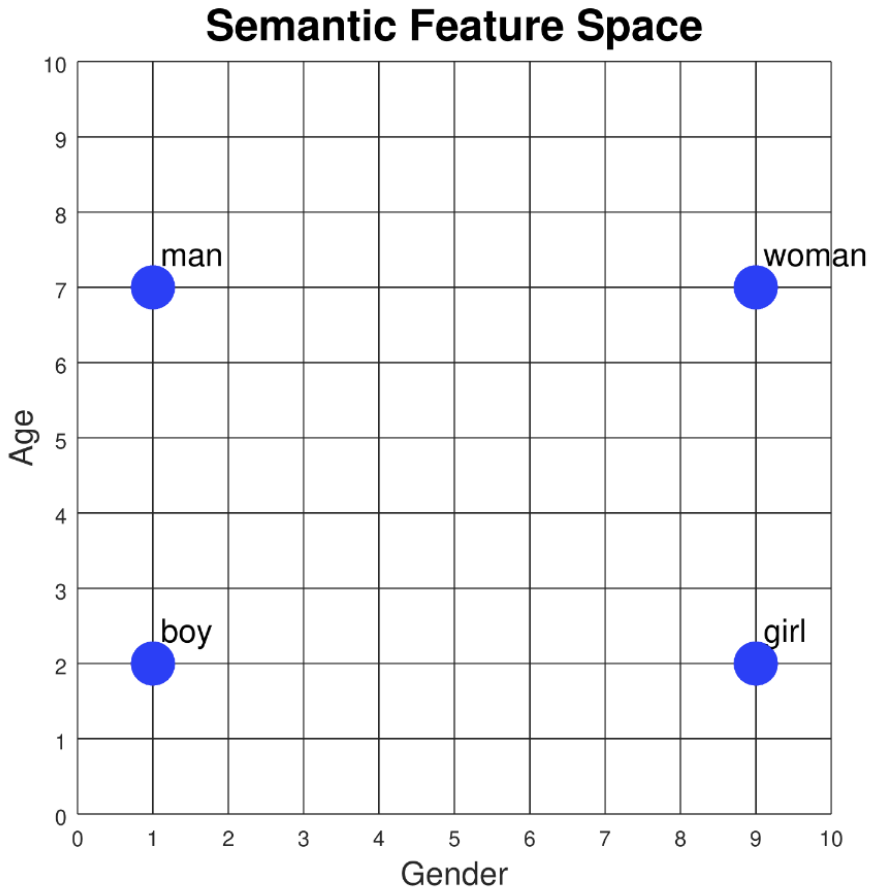
Some Math Basics

- Dot product
  - $a \bullet b = ||a||||b||\cos(\theta_{ab}) = a_1b_1+a_2b_2+ \ldots +a_nb_n$
- One can derive Cosine similarity
  - $\cos \theta_{ab} = a \bullet b \ / \ ||a||||b||$
- Softmax Function
  - If we take an input of [1,2,3,4,1,2,3], the softmax of that is [0.024, 0.064, 0.175, 0.475, 0.024, 0.064, 0.175].
  - The softmax function highlights the largest values and suppress other values, so that they are all positive and sum to 1.

Traditional Representation of a Word's Meaning

- How do we find the meaning of a word before Google
  - Dictionary: we have a word with unknown meaning→ we search the dictionary following some indices→ we find the work→ we get the meaning
  - not too useful in computational linguistic research
- WordNet
  - A large lexical database of English. Nouns, verbs, adjectives and adverbs are grouped into sets of cognitive synonyms (synsets), each expressing a distinct concept.
  - Depend on human labeling hence missing a lot, hard to automate this process.
- 1-hot vector
  - Hotel [0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0]
  - House [0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0]

Question 1

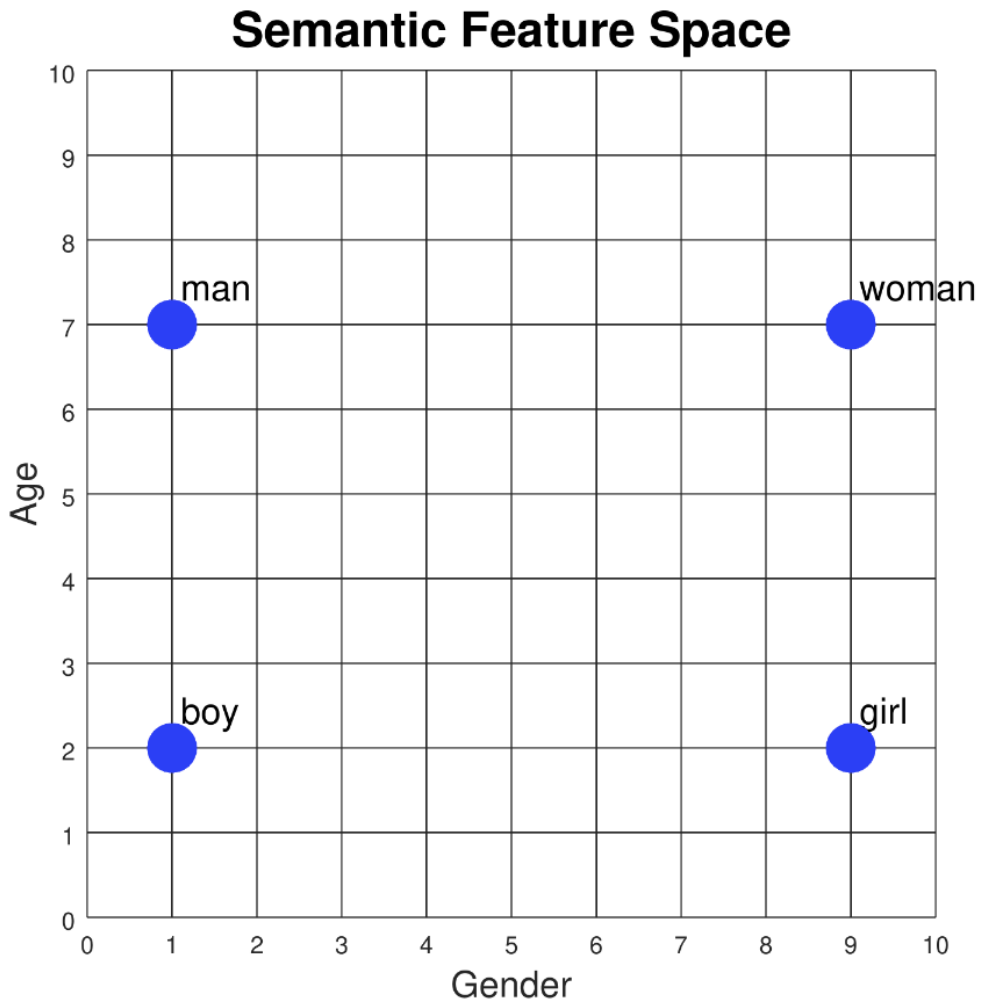


Assign Coordinate to each of the word

Answer

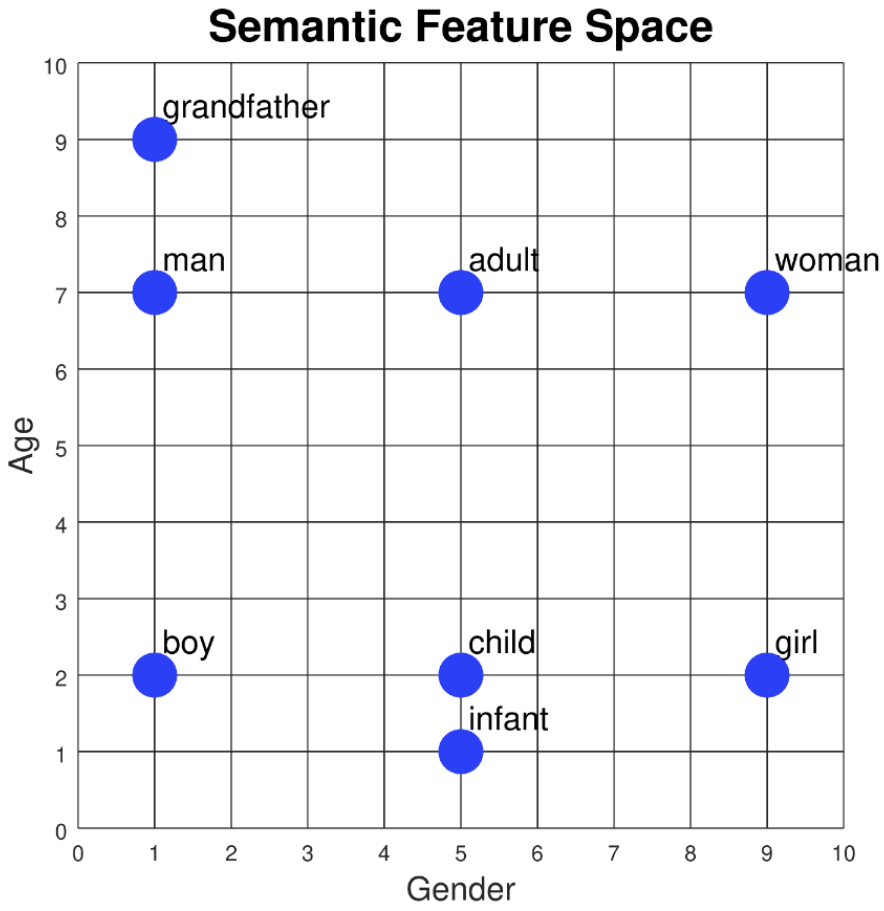
Word Coordinates				
		Gender	Age	
man	[	1,	7	]
woman	[	9,	7	]
boy	[	1,	2	]
girl	[	9,	2	]

Question 2



Quiz 2: How to Represent Adult, Child, Infant, Grandfather?

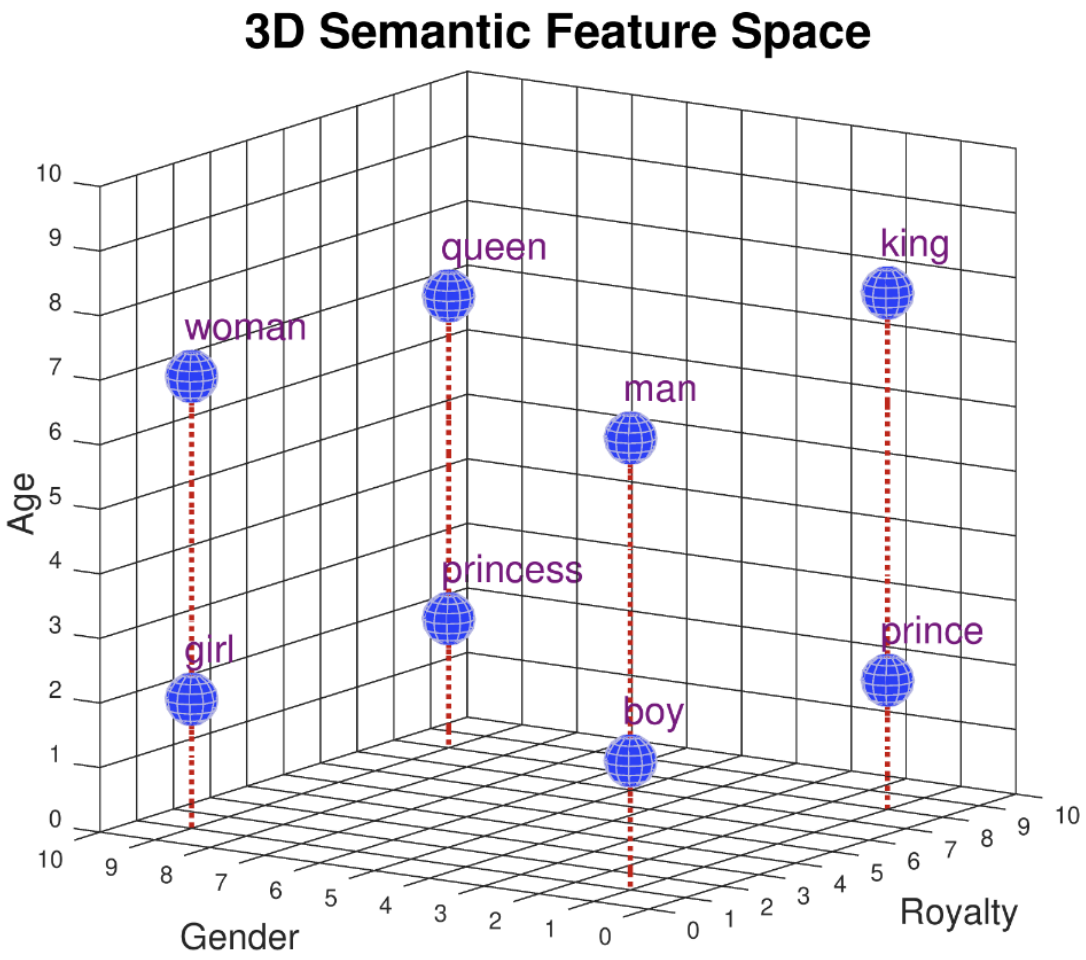
Answer



Word Coordinates

	Gender	Age
grandfather	[1,	9]
man	[1,	7]
adult	[5,	7]
woman	[9,	7]
boy	[1,	2]
child	[5,	2]
girl	[9,	2]
infant	[5,	1]

Now, how to Represent King, Queen, Prince and Princess?



Word Coordinates

	Gender	Age	Royalty
man	[1,	7,	1]
woman	[9,	7,	1]
boy	[1,	2,	1]
girl	[9,	2,	1]
king	[1,	8,	8]
queen	[9,	7,	8]
prince	[1,	2,	8]
princess	[9,	2,	8]

But, how to Represent Life, Cry, Coffee, Happiness or Marriage?

Representing Words by Their Context

- Distributional Hypothesis
  - Words that occur in similar context tend to have similar meanings
- The words "cat" and "dog" often appear in similar contexts, such as "I have a ___ as a pet," suggesting they share a similar meaning as animals or pets.
- "Bake" and "cook" tend to occur in sentences like "She loves to ___ cakes," indicating they are semantically related to food preparation.
- In the co ter" frequently appear ir ' showing their shared m

J.R.Firth 1957

- "You shall know a word by the company it keeps"
- One of the most successful ideas of modern statistical NLP!

Distributional Hypothesis

"tejuino"

C1: A bottle of ____ is on the table.

C2: Everybody likes ____.

C3: Don't have ____ before you drive.

C4: We make ____ out of corn.

Distributional Hypothesis

C1: A bottle of ____ is on the table.

C2: Everybody likes ____.

C3: Don't have ____ before you drive.

C4: We make ____ out of corn.

	C1	C2	C3	C4
tejuino	1	1	1	1
loud	0	0	0	0
motor-oil	1	0	0	0
tortillas	0	1	0	1
choices	0	1	0	0
wine	1	1	1	0

"words that occur in similar contexts tend to have similar meanings"

Tejuino is a traditional Mexican drink, especially popular in the states of **Jalisco, Nayarit, and Colima**.

🌽 **What It's Made Of**

Base ingredient: *Fermented corn dough* (masa de maíz), usually the same kind used for tortillas.

Sweetener: Piloncillo (unrefined cane sugar) or brown sugar.

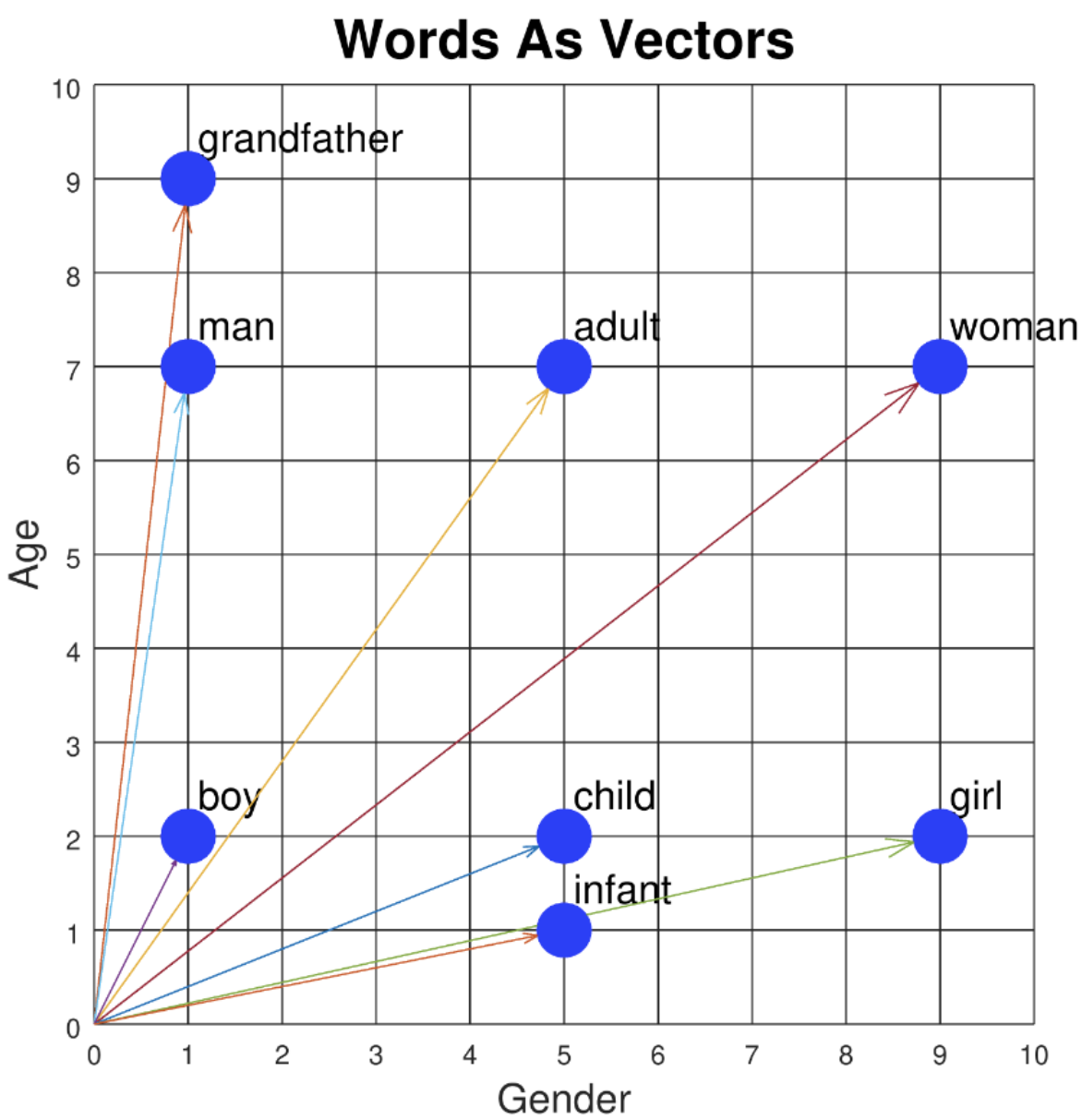
Flavoring: Lime juice and sometimes a pinch of salt.

Fermentation: It's lightly fermented, which gives it a tangy, slightly sour taste, but it's still considered a non-alcoholic or very low-alcohol beverage.

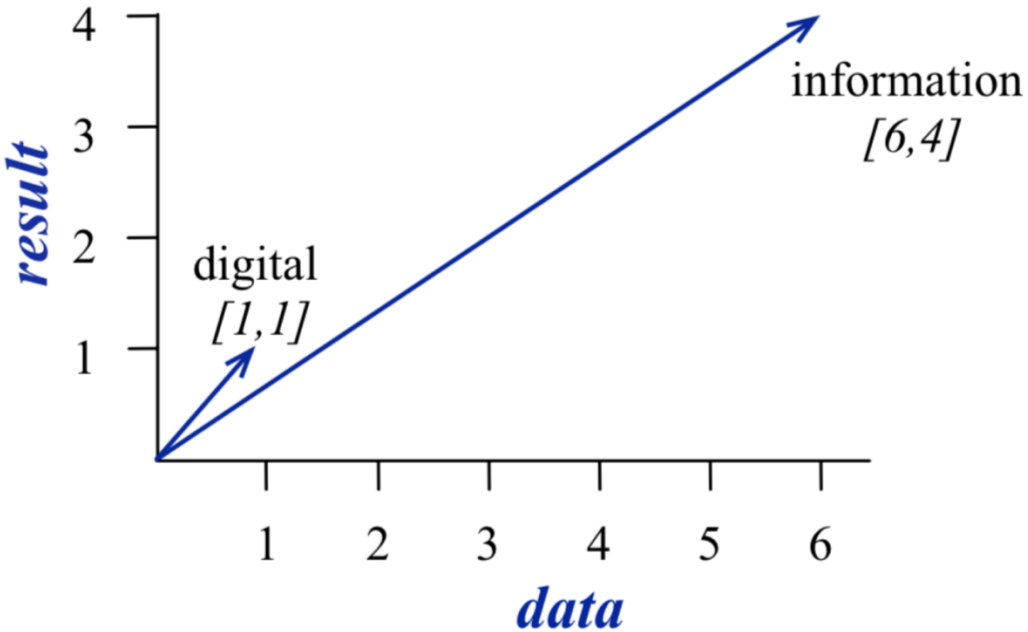
Word as Vectors

	berry	apricot	fruit	data	information	knowledge
berry	0	2	3	0	0	0
apricot	2	0	3	0	0	0
fruit	3	3	0	0	0	0
data	0	0	0	0	4	2
information	0	0	0	4	0	3
knowledge	0	0	0	2	3	0

- We'll build a new model of meaning focusing on similarity
  - Each word is a vector
  - Similar words are "nearby in space"
- A first solution
  - We can use context vectors to represent the meaning of words…
  - Word-word co-occurrence matrix


Words As Vectors
grandfather
man
adult
woman
boy
child
girl
infant
Age
Gender
0
1
2
3
4
5
6
7
8
9
10

Words as vectors



$$cos(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

$$cos(\mathbf{u}, \mathbf{v}) = \frac{\sum_{i=1}^{V} u_i v_i}{\sqrt{\sum_{i=1}^{V} u_i^2} \sqrt{\sum_{i=1}^{V} v_i^2}}$$

Assignment 1

- Hands on Assignment on a Benchmark or a hand craft dataset (+3 pts)
- Data Cleaning and preprocessing as you see needed
- Code (or jupyter notebook) + a short report